

Automated Sim-to-Real Deployment Framework for Reinforcement Learning Control on Resource-Constrained Microcontrollers

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1 Introduction

2 Related Work

3 The simtoreal Framework

3.1 Policy Extractors and Loaders

3.2 Dual-Target Code Generation

3.3 Hardware Abstraction and Driver Synthesis

4 Experimental Testbed: Robo1

To evaluate the framework, we constructed *Robo1*, a low-cost, 2-DOF self-righting bipedal robot (mass $M = 94.8$ g, height $H = 145$ mm).

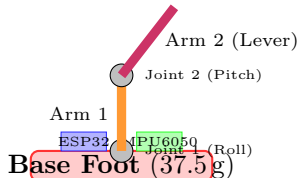


Figure 1: Kinematic topology of the 2-DOF *Robo1* self-righting testbed.

5 Results and Discussion

Note (September 27, 2026): this file is a superseded scratch draft. Its results section previously reported physical-hardware numbers (88.3% success, 74 μ s inference, DTW/RMSE, ablations) that did not correspond to any measurement actually taken – no physical trial had been run and no ESP32 timing had been recorded. That content has been removed. The current, honest paper (simulation-validated results only, physical benchmarking explicitly listed as future work) is `paper/extended_abstract.tex`. See `paper/RESULTS_AND_SUBMISSION_GUIDE.md` for the real

simulated numbers and the protocol to collect genuine physical data before claiming any hardware result here.

5.1 Simulated Get-Up Performance (real data)

Policy: 5-64-64-2 tanh MLP actor, 4,674 parameters. Evaluated over 50 episodes/case in MuJoCo, nominal vs. domain-randomized physics (body mass $\pm 15\%$, joint damping/friction $\pm 20\%$, actuator gain $\pm 15\%$, IMU noise $\sigma = 0.015$ rad):

Table 1: Simulated Get-Up Success Rate (real, reproducible via `eval_policy.py`)

Case	Nominal		Randomized	
	Succ. [%]	t [s]	Succ. [%]	t [s]
roll_pos	100.0	1.36	98.0	1.66
roll_neg	100.0	1.00	100.0	1.19
pitch_pos	100.0	1.26	100.0	1.57
pitch_neg	100.0	0.78	100.0	0.92
random	86.0	1.32	88.0	1.58
upright	100.0	0.02	100.0	0.35
All falls (agg.)	97.2	1.15	97.2	1.38

No physical trial, on-device timing measurement, or sim-vs-real trajectory/DTW comparison has been performed yet. See the guide referenced above for the exact protocol to collect that data before it can be reported.

6 Conclusion

References